

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 10

Diameter Protocol

پروتکل Diameter



CLIENT

AGENT

SERVER

HEADER · AVP · AVP · AVP



Lecture Notes

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1 Why Diameter Was Introduced

RADIUS Limitations

RADIUS (Remote Authentication Dial-In User Service) was designed in the 1990s for dial-up access. As networks evolved toward 3G/4G, RADIUS could not meet modern requirements:

Limitation	Impact
UDP transport (unreliable)	Messages can be lost silently; no built-in retransmission
No server-initiated messages	Server cannot push policy changes or disconnect users proactively
Limited AVP space	8-bit attribute type field = max 255 attribute types
No application-layer ACK	Cannot confirm message delivery at application level
Weak security model	Shared secret + MD5 only; no hop-by-hop TLS built-in
No capability negotiation	Peers cannot discover supported features
Client-server only	No peer-to-peer; no agent/relay architecture

Diameter's Answer

Diameter (the name implies “twice the RADIUS”) was defined in **RFC 6733** to address all these limitations:

- Reliable transport (TCP/SCTP)

- Server-initiated messages (push model)
- 32-bit AVP code space (4+ billion attribute types)
- Built-in capability exchange and error handling
- Agent architecture for scalable routing
- Native TLS/DTLS support

⚠ **Key Concept:** Diameter is a **CONTROL PLANE** protocol only. It carries signaling (authentication, authorization, accounting, policy) — never user data traffic.

2 Diameter vs RADIUS Comparison

Feature	RADIUS	Diameter
RFC	RFC 2865/2866	RFC 6733
Transport	UDP (unreliable)	TCP/SCTP (reliable)
Port	1812/1813	3868
Connection	Per-request	Persistent peer connections
Direction	Client→Server only	Peer-to-peer (both can initiate)
AVP code space	8-bit (255 max)	32-bit (4 billion+)
Security	Shared secret + MD5	TLS/DTLS, IPsec
Max message size	4096 bytes	16 MB (24-bit length)
Failover	Application-dependent	Built-in (transport-level detection)
Capability negotiation	None	CER/CEA exchange
Server-initiated	No	Yes (e.g., RAR, ASR)
Agent support	Limited (proxy)	Relay, Proxy, Redirect, Translation
Accounting	Interim updates only	Real-time with guaranteed delivery
Vendor AVPs	Limited	Native vendor-ID support
Used in	Wi-Fi, VPN, ISP	LTE/4G core, IMS, VoLTE

3 Diameter Architecture

3.1 Diameter Nodes

Node Type	Role
Diameter Client	Generates requests (e.g., MME, P-GW)
Diameter Server	Processes requests, returns answers (e.g., HSS, PCRF, OCS)
Diameter Agent	Intermediary that routes/processes messages

■ Agent Types

Agent	Function
Relay	Routes messages based on realm; does NOT modify AVPs
Proxy	Routes messages AND may modify AVPs (policy enforcement)
Redirect	Returns routing information to sender (does not forward)
Translation	Translates between Diameter and other protocols (e.g., RADIUS↔Diameter)

3.2 Peer Connections

- **Transport:** TCP (default) or SCTP (preferred for multi-homing)
- **Port:** 3868 (plaintext) or 5868 (TLS)
- **Persistent:** Connection stays up; monitored by watchdog (DWR/DWA)
- **Peer-to-peer:** Both endpoints can initiate messages
- **Capability Exchange:** CER/CEA must succeed before application messages

3.3 Realm-Based Routing

Diameter routes messages based on **realms** (domain-like identifiers):

- Each node is configured with a realm (e.g., `epc.mnc001.mcc208.3gppnetwork.org`)
- Routing table maps realms → peer connections
- Agents use **Destination-Realm** AVP to determine next hop
- Enables multi-operator and roaming scenarios

4 Diameter Message Structure

4.1 Message Header (20 bytes fixed)

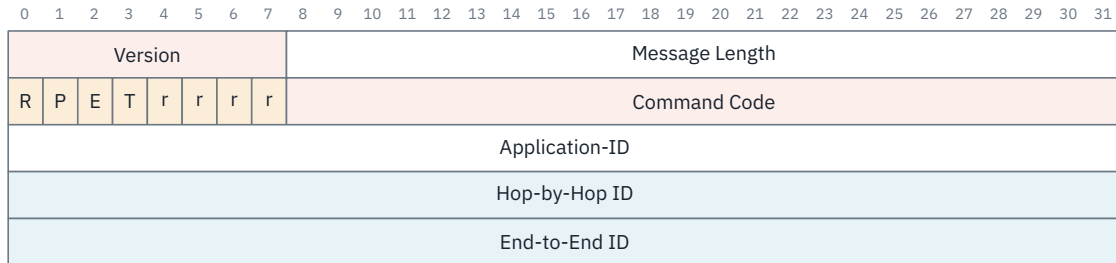


Figure 10.1 Diameter message header (20 bytes)

Field	Size	Description
Version	1 byte	Always 1
Message Length	3 bytes	Total message length including header
Flags	1 byte	R=Request, P=Proxiable, E=Error, T=Retransmit
Command Code	3 bytes	Identifies the command (e.g., 318 = AIR)
Application-ID	4 bytes	Identifies the application (e.g., 16777251 = S6a)
Hop-by-Hop ID	4 bytes	Matches request/answer on a single hop (unique per connection)
End-to-End ID	4 bytes	Matches request/answer end-to-end (unique globally)

■ Header Flags

Flag	Bit	Meaning
R	0	Request (1) or Answer (0)
P	1	Proxiable — can be relayed/proxied
E	2	Error — answer contains error
T	3	Retransmitted message

4.2 AVP (Attribute-Value Pair) Structure

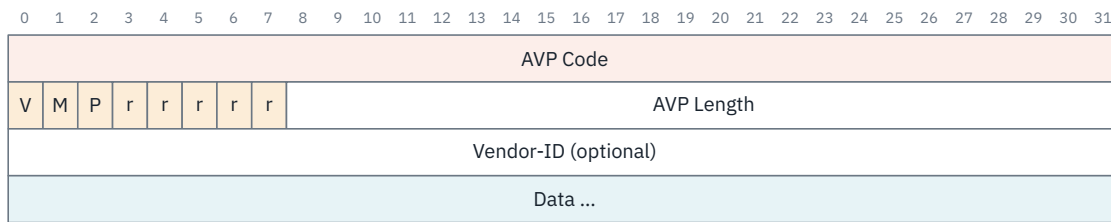


Figure 10.2 Diameter AVP structure

Field	Description
AVP Code	32-bit identifier for the attribute
V flag	Vendor-specific — Vendor-ID field is present
M flag	Mandatory — receiver MUST understand this AVP
P flag	Protected — end-to-end encryption needed
AVP Length	Total AVP length (header + data)
Vendor-ID	32-bit vendor identifier (present only if V=1)
Data	The actual value (various types)

■ AVP Data Types

Type	Description	Example
OctetString	Raw bytes	RAND, AUTN
UTF8String	Text string	User-Name
Unsigned32	32-bit unsigned integer	Result-Code
Unsigned64	64-bit unsigned integer	Accounting counters
Address	IP address (IPv4/IPv6)	Framed-IP-Address
Time	Seconds since Jan 1, 1900	Event-Timestamp
Enumerated	Named integer values	Auth-Session-State
Grouped	Contains other AVPs	Subscription-Data

4.3 Grouped AVPs

Grouped AVPs contain other AVPs as their data — enabling hierarchical structures:

```

Subscription-Data (Grouped AVP)
├─ MSISDN: +33612345678
├─ Access-Restriction-Data: 0x00000000
├─ Subscriber-Status: SERVICE_GRANTED
├─ APN-Configuration-Profile (Grouped)
│   └─ Context-Identifier: 1
│       └─ APN-Configuration (Grouped)
│           └─ Service-Selection: "internet"
│               └─ PDN-Type: IPv4v6
│                   └─ EPS-Subscribed-QoS-Profile (Grouped)
│                       └─ QoS-Class-Identifier: 9
│                           └─ Allocation-Retention-Priority (Grouped)
│                               └─ Priority-Level: 15
│                                   └─ Pre-emption-Capability: NOT_PRE_EMPTY

```

Key AVPs Table

AVP Name	Code	Type	Used In
Session-Id	263	UTF8String	All messages
Origin-Host	264	DiameterIdentity	All messages
Origin-Realm	296	DiameterIdentity	All messages
Destination-Host	293	DiameterIdentity	Requests
Destination-Realm	283	DiameterIdentity	Requests
Result-Code	268	Unsigned32	All answers
Auth-Session-State	277	Enumerated	Auth messages
User-Name	1	UTF8String	User identification (IMSI)
Visited-PLMN-Id	1407	OctetString	S6a messages
Subscription-Data	1400	Grouped	S6a (ULA)
Charging-Rule-Install	1001	Grouped	Gx (CCA)
CC-Request-Type	416	Enumerated	Gy/Gx (CCR)

5 Diameter Commands

5.1 Naming Convention

All Diameter commands follow the pattern:

- **Request:** *X-Request* (abbreviated *XR*) — R flag = 1
- **Answer:** *X-Answer* (abbreviated *XA*) — R flag = 0

Every request MUST receive exactly one answer.

5.2 Base Protocol Commands (Application-ID = 0)

Command	Code	Abbreviation	Purpose
Capabilities-Exchange-Request	257	CER	Initiate peer connection, exchange capabilities
Capabilities-Exchange-Answer	257	CEA	Respond with own capabilities
Device-Watchdog-Request	280	DWR	Heartbeat — verify peer is alive
Device-Watchdog-Answer	280	DWA	Confirm alive status
Disconnect-Peer-Request	282	DPR	Gracefully close connection
Disconnect-Peer-Answer	282	DPA	Acknowledge disconnection
Accounting-Request	271	ACR	Send accounting record
Accounting-Answer	271	ACA	Acknowledge accounting record
Abort-Session-Request	274	ASR	Server requests session termination
Abort-Session-Answer	274	ASA	Client acknowledges termination
Re-Auth-Request	258	RAR	Server requests re-authentication
Re-Auth-Answer	258	RAA	Client acknowledges re-auth

5.3 Diameter Message Exchange Flow

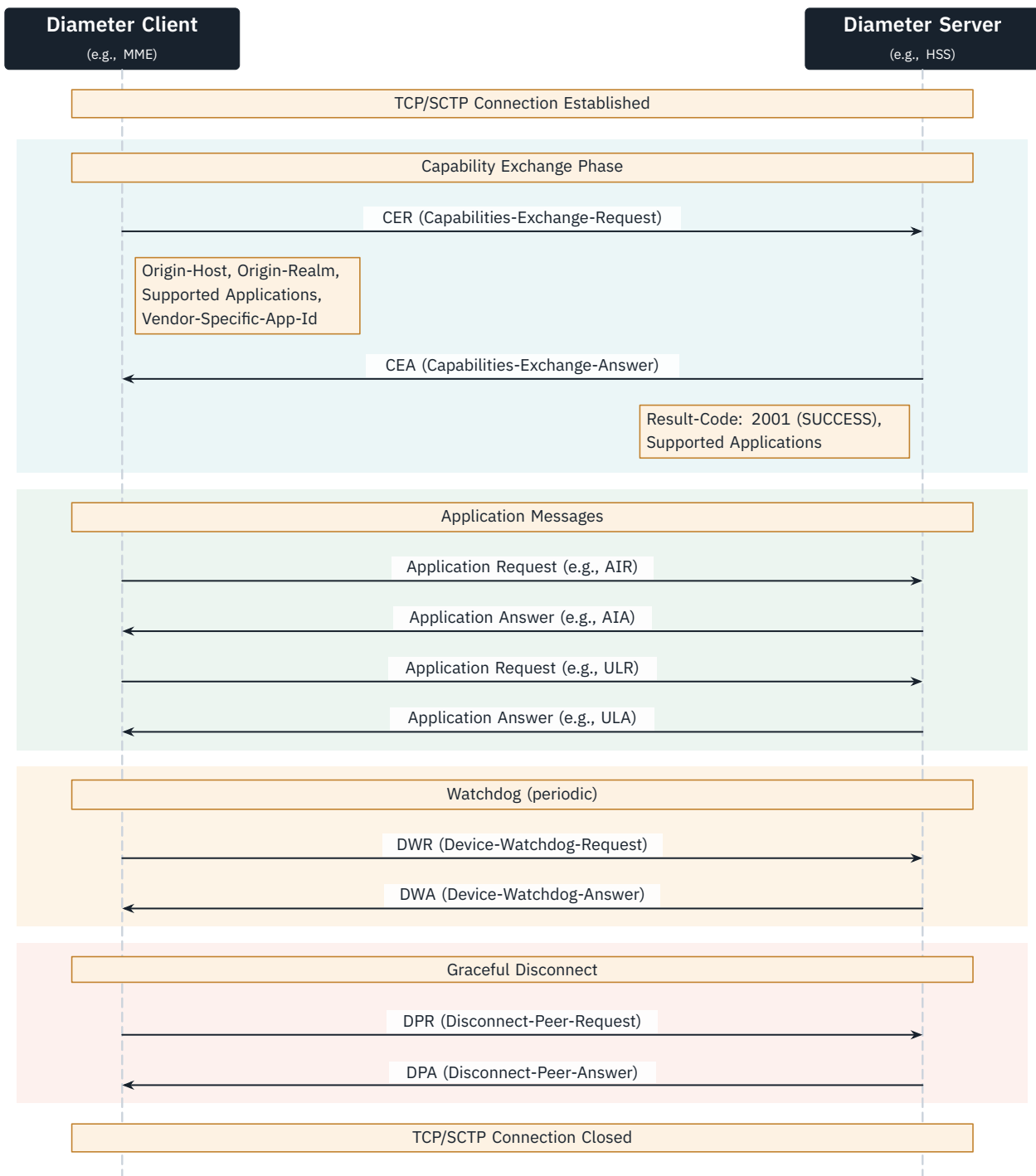


Figure 10.3 Diameter peer life cycle: capability exchange, application messages, watchdog and disconnect

5.4 Result Codes

Code	Name	Meaning
2001	DIAMETER_SUCCESS	Request processed successfully
3xxx	Protocol Errors	Redirect, unable to deliver
4xxx	Transient Failures	Retry may succeed (e.g., 4012 = DIAMETER_UNABLE_TO_COMPLY)
5xxx	Permanent Failures	Do not retry (e.g., 5001 = DIAMETER_AVP_UNSUPPORTED)
5004	DIAMETER_UNKNOWN_SESSION_ID	Session not found
5012	DIAMETER_UNABLE_TO_COMPLY	General permanent failure

6 Important Diameter Applications in LTE

6.1 Overview of LTE Diameter Interfaces

Interface	Endpoints	Application-ID	Purpose
S6a	MME ↔ HSS	16777251	Authentication, subscriber data
Gx	PCRF ↔ P-GW	16777238	Policy and Charging Control (PCC)
Gy	OCS ↔ P-GW	4 (Credit-Control)	Online charging
Rx	AF ↔ PCRF	16777236	Application-level QoS/media authorization
S13	MME ↔ EIR	16777252	Equipment identity check (IMEI)
S6d	SGSN ↔ HSS	16777251	2G/3G authentication via Diameter
SWx	3GPP AAA ↔ HSS	16777265	Non-3GPP access authentication
Sh	AS ↔ HSS	16777217	IMS service data

6.2 S6a Interface (MME ↔ HSS)

The most critical Diameter interface in LTE — handles all subscriber authentication and data management.

Command	Code	Direction	Purpose
Authentication-Information-Request	318	MME→HSS	Request authentication vectors
Authentication-Information-Answer	318	HSS→MME	Return auth vectors (RAND, AUTN, XRES, K_{ASME})
Update-Location-Request	316	MME→HSS	Register UE location at MME
Update-Location-Answer	316	HSS→MME	Return subscription data
Purge-UE-Request	321	MME→HSS	Inform HSS that UE data purged
Purge-UE-Answer	321	HSS→MME	Acknowledge purge
Cancel-Location-Request	317	HSS→MME	HSS tells MME to detach UE
Cancel-Location-Answer	317	MME→HSS	Acknowledge cancel
Notify-Request	323	MME→HSS	Notify HSS of terminal info
Notify-Answer	323	HSS→MME	Acknowledge notification

6.3 Gx Interface (PCRF ↔ P-GW)

Controls **Policy and Charging Control (PCC)** — decides what QoS and charging rules apply to each session.

Command	Code	Direction	Purpose
CC-Request (Initial)	272	P-GW→PCRF	New IP-CAN session — request PCC rules
CC-Answer (Initial)	272	PCRF→P-GW	Install PCC rules (QoS, gates, charging)
CC-Request (Update)	272	P-GW→PCRF	Session modification (e.g., new bearer)
CC-Answer (Update)	272	PCRF→P-GW	Updated PCC rules
CC-Request (Termination)	272	P-GW→PCRF	Session ended
CC-Answer (Termination)	272	PCRF→P-GW	Acknowledge termination
Re-Auth-Request	258	PCRF→P-GW	Push new PCC rules mid-session
Re-Auth-Answer	258	P-GW→PCRF	Acknowledge rule update

CC-Request-Type values: INITIAL(1), UPDATE(2), TERMINATION(3), EVENT(4)

6.4 Gy Interface (OCS ↔ P-GW)

Handles **online charging** — credit reservation and debit in real-time.

Command	Code	Direction	Purpose
CC-Request (Initial)	272	P-GW→OCS	Reserve initial credit quota
CC-Answer (Initial)	272	OCS→P-GW	Grant credit (units/time/volume)
CC-Request (Update)	272	P-GW→OCS	Report usage, request more credit
CC-Answer (Update)	272	OCS→P-GW	Grant additional credit or deny
CC-Request (Termination)	272	P-GW→OCS	Final report, release reservation
CC-Answer (Termination)	272	OCS→P-GW	Acknowledge final

Gx vs Gy: **Gx** = policy rules (what QoS to apply), **Gy** = charging (how much credit remains)

6.5 Rx Interface (AF ↔ PCRF)

Allows application functions (e.g., IMS P-CSCF for VoLTE) to request specific QoS from the network.

Command	Code	Direction	Purpose
AA-Request	265	AF→PCRF	Request media authorization (SDP info)
AA-Answer	265	PCRF→AF	Confirm media session authorized
Session-Termination-Request	275	AF→PCRF	Media session ended
Session-Termination-Answer	275	PCRF→AF	Acknowledge
Re-Auth-Request	258	PCRF→AF	Notify AF of bearer changes
Abort-Session-Request	274	PCRF→AF	PCRF aborts media session

6.6 S13 Interface (MME ↔ EIR)

Equipment identity check — validates IMEI against blacklist/greylist.

Command	Code	Direction	Purpose
ME-Identity-Check-Request	324	MME→EIR	Send IMEI for validation
ME-Identity-Check-Answer	324	EIR→MME	Return equipment status (white/grey/black)

7 Diameter in LTE Authentication (S6a Deep Dive)

7.1 Authentication Flow

When a UE attaches to the LTE network, the MME uses S6a to:

1. **Obtain authentication vectors** from the HSS (AIR/AIA)
2. **Register UE location** at the MME (ULR/ULA)

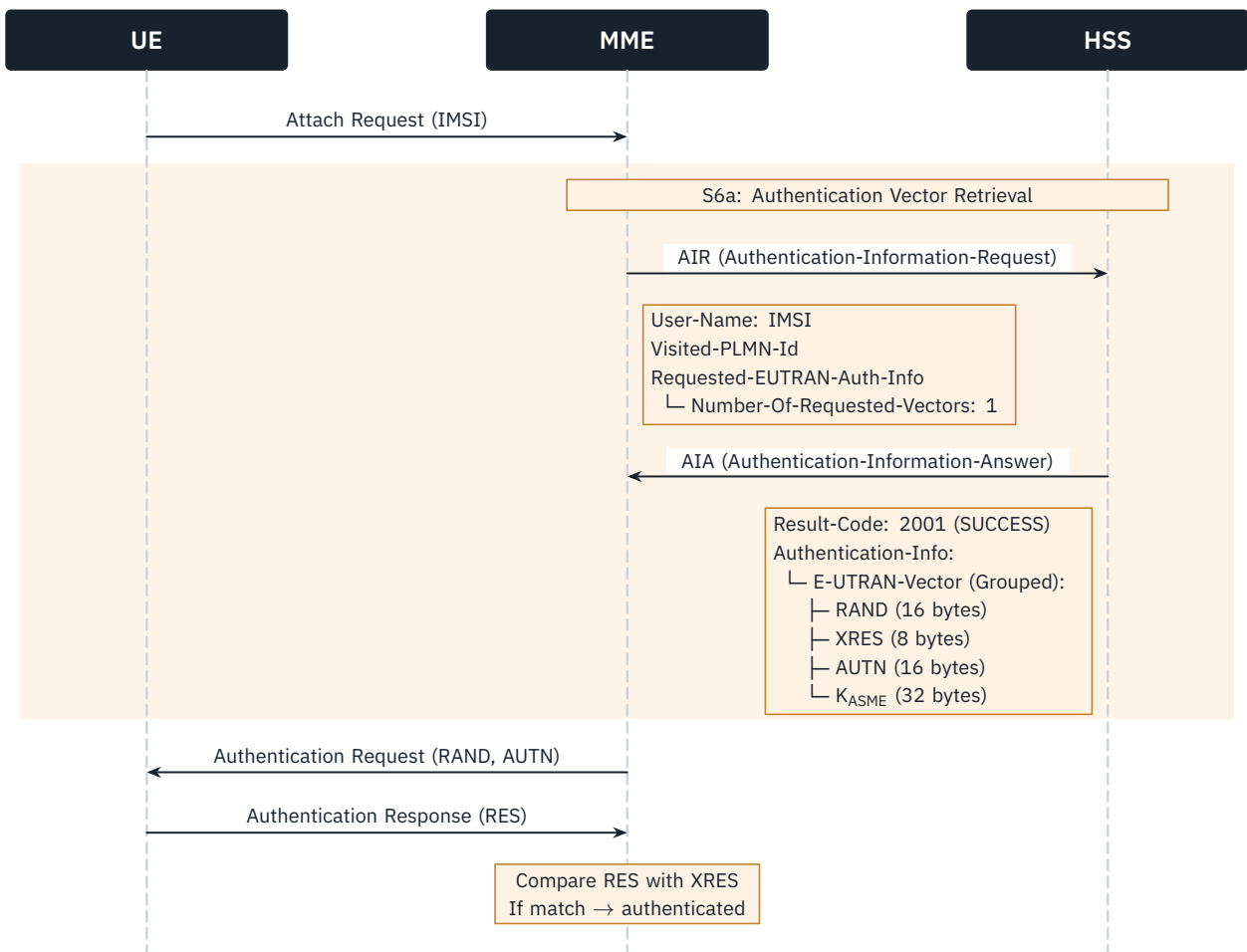


Figure 10.4 S6a messages and their AVPs during attach (part 1 of 2)

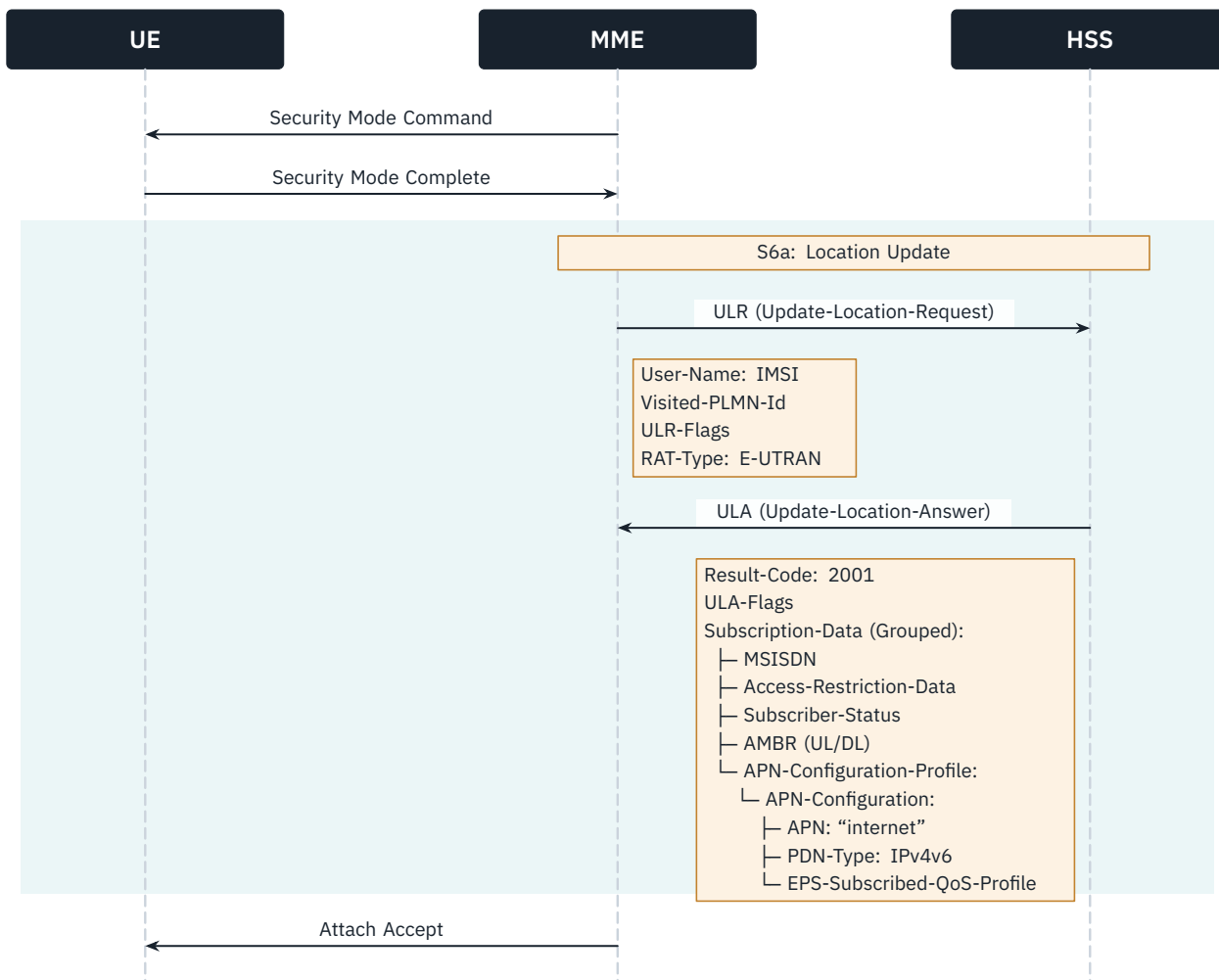


Figure 10.5 S6a messages and their AVPs during attach (part 2 of 2)

7.2 AIR (Authentication-Information-Request) Details

Purpose: MME requests fresh authentication vectors from HSS

Key AVPs in AIR:

- | AVP | Purpose |
- |---|----|
- | User-Name | IMSI of the subscriber |
- | Visited-PLMN-Id | 3-byte MCC+MNC of visited network |
- | Requested-EUTRAN-Authentication-Info | Specifies how many vectors needed |
- | Number-Of-Requested-Vectors | Typically 1-5 |
- | Re-Synchronization-Info | Sent if previous auth failed (SQN sync) |

7.3 AIA (Authentication-Information-Answer) Details

Purpose: HSS returns E-UTRAN authentication vectors

Key AVPs in AIA:

- | AVP | Purpose |
- |---|----|
- | Result-Code | 2001=success, 5001=user unknown |

- | Authentication-Info (Grouped) | Contains the vectors |
- | E-UTRAN-Vector (Grouped) | One complete auth vector |
- | └─ RAND | 128-bit random challenge |
- | └─ XRES | Expected Response (for verification) |
- | └─ AUTN | Authentication Token (for mutual auth) |
- | └─ K_{ASME} | Key for NAS/AS security derivation |

7.4 ULR (Update-Location-Request) Details

Purpose: MME registers itself as serving MME for this subscriber

Key AVPs in ULR:

- | AVP | Purpose |
- | ---|----|
- | User-Name | IMSI |
- | Visited-PLMN-Id | Serving network identity |
- | RAT-Type | E-UTRAN (LTE) |
- | ULR-Flags | Indicates initial attach vs TAU |

7.5 ULA (Update-Location-Answer) Details

Purpose: HSS acknowledges and returns complete subscription profile

Key AVPs in ULA:

- | AVP | Purpose |
- | ---|----|
- | Result-Code | Success or failure |
- | Subscription-Data (Grouped) | Complete subscriber profile |
- | └─ MSISDN | Phone number |
- | └─ AMBR | Aggregate Max Bit Rate (UL + DL) |
- | └─ APN-Configuration-Profile | All configured APNs |
- | └─ Subscriber-Status | SERVICE_GRANTED or BARRED |

Gx Policy Flow (CCR-I/CCA-I at Session Start)

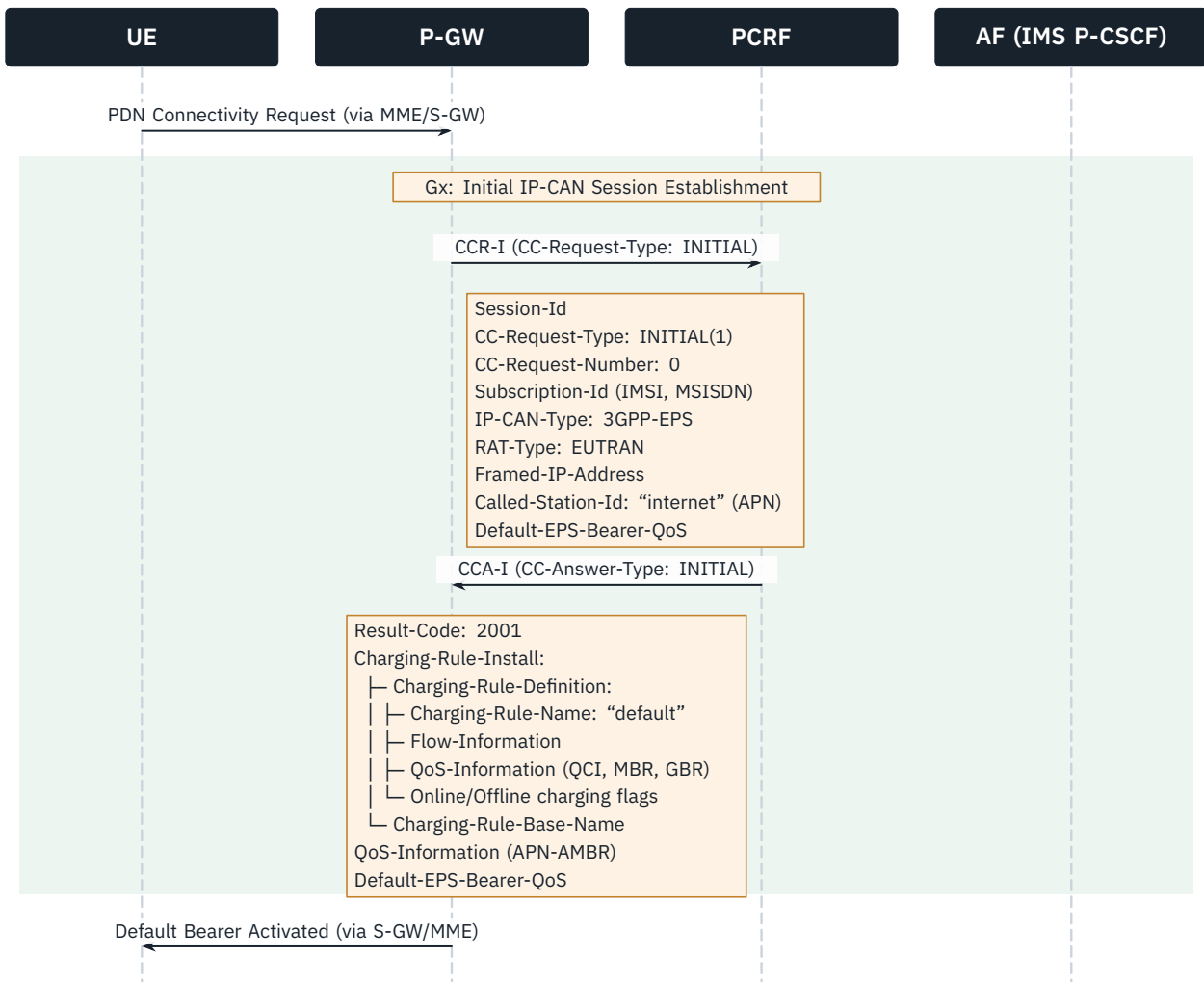


Figure 10.6 Gx and Rx policy flow for a default bearer and a VoLTE call (part 1 of 2)

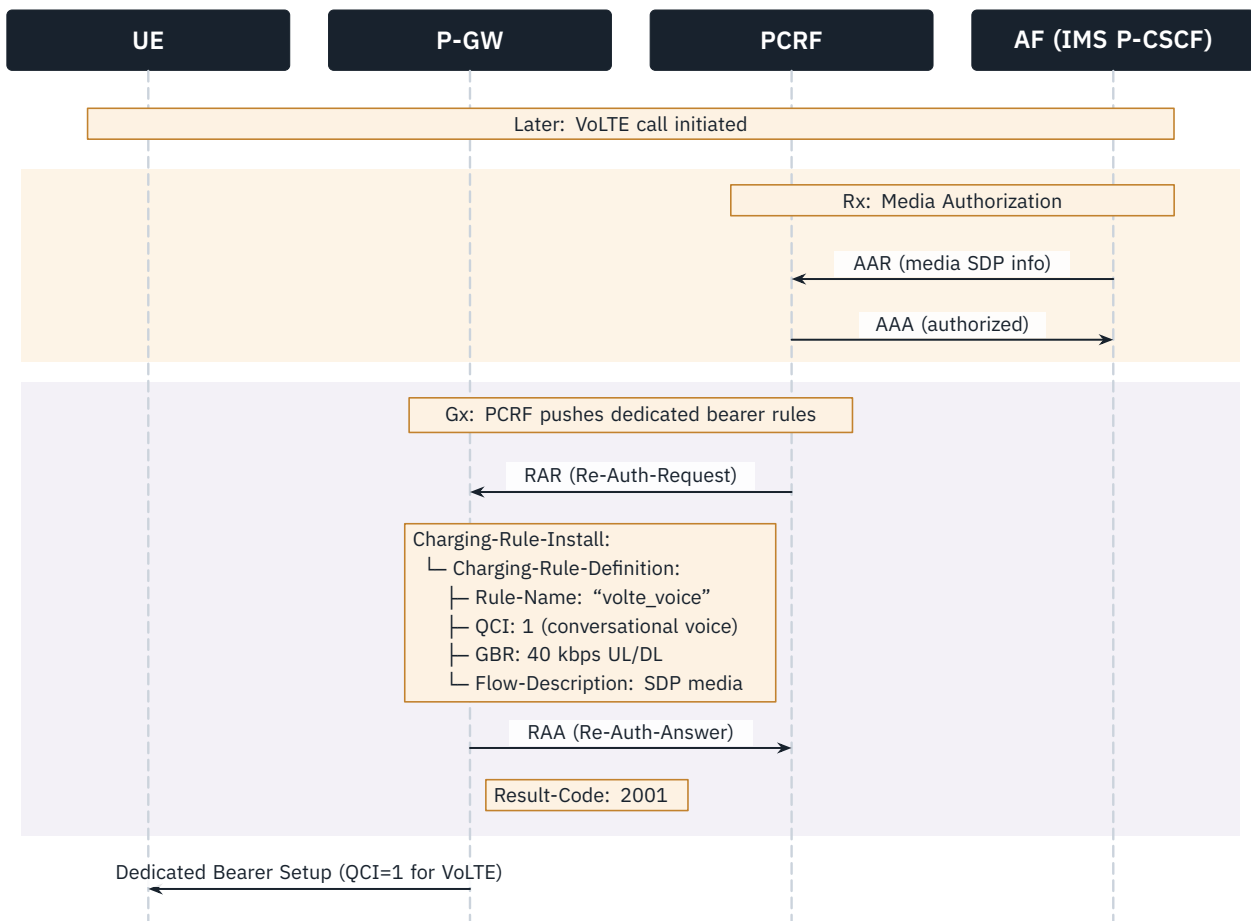


Figure 10.7 Gx and Rx policy flow for a default bearer and a VoLTE call (part 2 of 2)

8 Diameter vs HTTP/2 in 5G

8.1 The Transition

Aspect	4G (Diameter)	5G (HTTP/2 + JSON)
Protocol	Diameter (binary, TCP/SCTP)	HTTP/2 (text-based headers, binary framing)
Data format	AVPs (binary TLV)	JSON (human-readable)
Architecture	Point-to-point peers	Service-Based Interface (SBI)
Discovery	Static peer config / DRA	NRF (Network Repository Function)
Interface naming	S6a, Gx, Gy, Rx	Nausf, Npcf, Nchf, Naf
Session state	Stateful (session-id maintained)	Stateless (REST-like)
Connection model	Persistent peer connections	HTTP request/response
Scaling	Vertical (bigger peers)	Horizontal (cloud-native, k8s)
Message routing	Diameter Routing Agents (DRA)	Service Mesh / API Gateway
Standard	IETF RFC 6733 + 3GPP	3GPP + IETF HTTP/2 (RFC 7540)

8.2 Why 5G Moved Away from Diameter

1. **Simpler tooling:** HTTP/2 + JSON leverages existing web infrastructure (proxies, load balancers, API gateways)
2. **Web-native:** Standard REST APIs — easier for developers, standard debugging tools
3. **Stateless design:** Enables cloud-native microservices, horizontal scaling, Kubernetes orchestration
4. **Cloud-friendly:** No persistent peer connections; standard load balancing works
5. **Flexibility:** JSON schema evolution is easier than binary AVP definitions
6. **Ecosystem:** Massive HTTP/2 ecosystem (libraries, tools, monitoring)

8.3 Diameter Equivalent Functions in 5G

4G Diameter Interface	5G SBI Equivalent	NF Involved
S6a (MME↔HSS)	Nudm (UDM services)	AMF ↔ UDM/AUSF
Gx (PCRF↔P-GW)	Npcf (Policy services)	SMF ↔ PCF
Gy (OCS↔P-GW)	Nchf (Charging services)	SMF ↔ CHF
Rx (AF↔PCRF)	Npcf + Naf	AF ↔ PCF (via NEF)

8.4 Migration: 4G/5G Interworking

Diameter is NOT gone — it persists at interworking boundaries:

- **Diameter Routing Agent (DRA)** still used in 4G
- **Interworking Function (IWF)** translates Diameter ↔ HTTP/2 at 4G/5G border
- NSA (Non-Standalone) 5G still uses EPC with Diameter
- Full migration to SA (Standalone) 5G eliminates Diameter within the core

9 Diameter = Control Plane Only

■ Critical Distinction

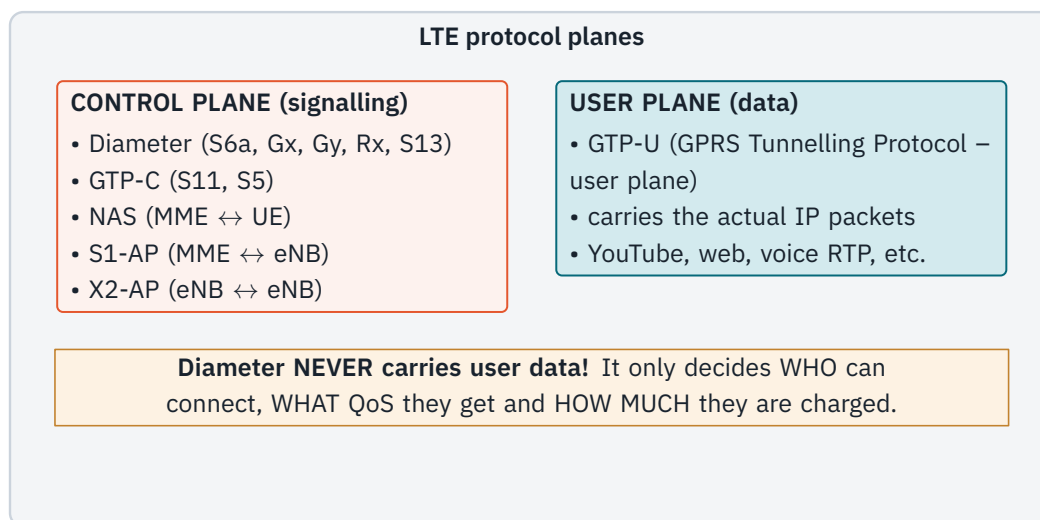


Figure 10.8 Diameter belongs to the control plane only

What Diameter Controls vs What Carries Data

Function	Protocol	Plane
Authenticate subscriber	Diameter S6a	Control
Authorize QoS policy	Diameter Gx	Control
Check credit balance	Diameter Gy	Control
Authorize media session	Diameter Rx	Control
Validate IMEI	Diameter S13	Control
Carry YouTube video	GTP-U	User
Carry VoLTE voice (RTP)	GTP-U	User
Carry web browsing	GTP-U	User

★ **Remember:** Diameter tells the network *what to do*. GTP-U *does it*.

Summary

Topic	Key Takeaway
Why Diameter	RADIUS too limited for mobile core (no reliable transport, no push, limited AVPs)
Architecture	Client/Server/Agent model with persistent TCP/SCTP peers and realm routing
Messages	20-byte header + AVPs; every Request gets exactly one Answer
S6a	Authentication (AIR/AIA) + Location Update (ULR/ULA) — the backbone of LTE mobility
Gx	Policy control — installs QoS rules on bearers (CCR/CCA + RAR/RAA)
Gy	Online charging — real-time credit control
Rx	Application QoS — IMS/VoLTE media authorization
5G transition	HTTP/2 + JSON replaces Diameter; IWF for interworking
Scope	Control plane ONLY — never carries user data

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